Writing a Java Program (2/2)

EXTRA MATERIAL

** using our new programming skills to write a real program (and learning some new ones on the way!)

** **`ArrayList`**

** Java API (brief introduction)

Chapter 6 (*) – “Head First Java” book

These slides are left as practice and self-study.

(*) *Example from the chapter*

Building the real "Sink a Dot Com" (Recap only)

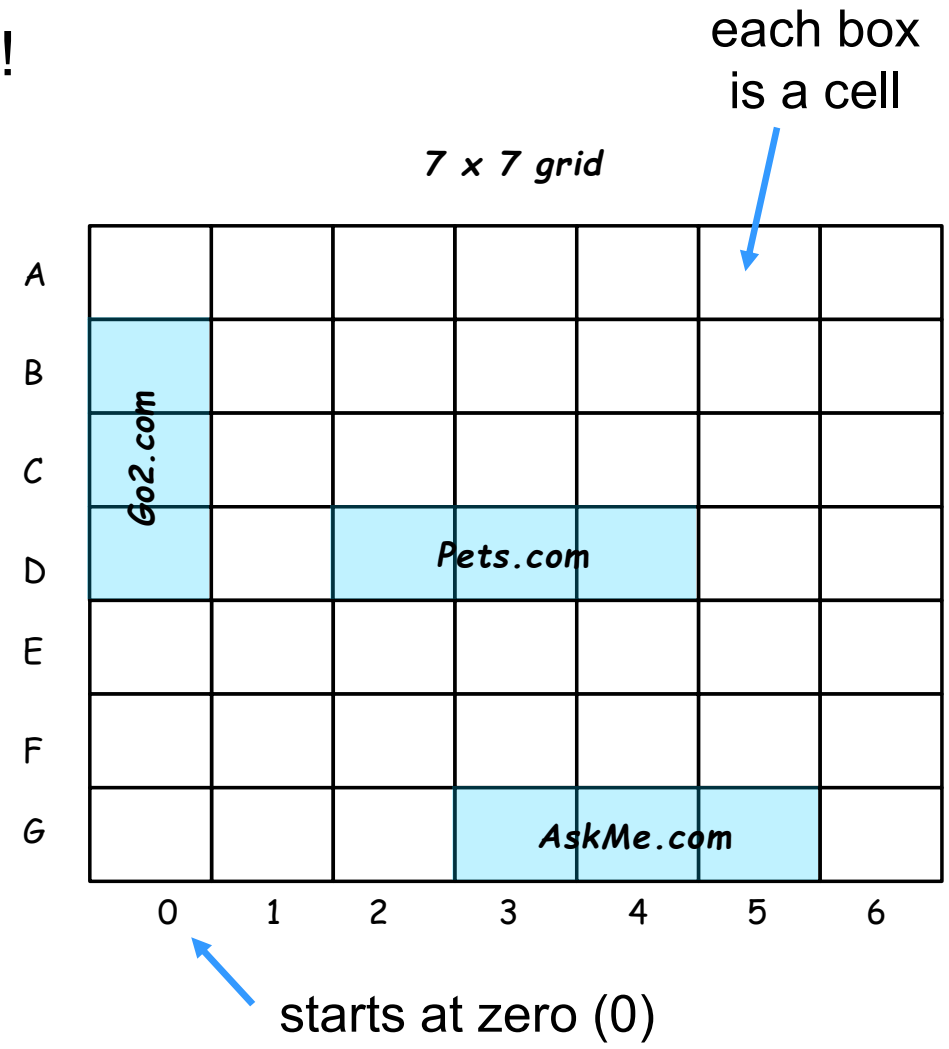
- We have been writing the simple version of the "Dot Com" game.
- Now we have to build the whole thing!

GOAL

- Sink all the computer's "Dot Coms" in the fewest number of guesses.
- You are given a rating, based on how well you perform.

SETUP

- A virtual 7x7 board with 3 randomly placed "Dot Coms".
- After that, the player should be prompted to enter their first guess.



What needs to change?

1. **DotCom** class
   - a. Needs a name variable! Remember that the dot com needs to be able to print its name after being killed! (Ouch! You sunk Pets.com ☹)

2. **DotComBust** class (the game)
   - a. Need 3 **DotComs** instead of 1!
   - B. Give each of the **DotCom**s a name when created! Need to use a *setter* to do it!
   - C. Put the **DotCom**s on a grid, rather than a single row. This is a bit complicated so:

to the rescue! Put this in the mysterious **GameHelper** class.

   - D. Check the user's guess with ALL 3 of the **DotCom**s!
   - e. Keep playing until ALL 3 **DotCom**s are killed!
   - F. Get out of **main()**.

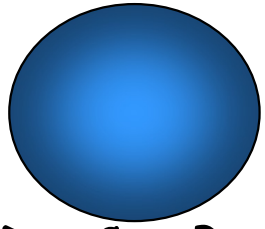
The classes

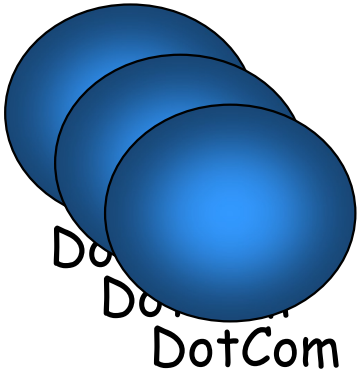
DotCom	DotComBust	GameHelper
The actual DotCom objects. DotComs know their *name*, *location* and how to check a user guess for a match.	**The game class.** Makes DotComs and gets user input, plays until all DotComs are dead	**The helper class.** *Ready bake...* Can accept user command-line input and make DotCom locations

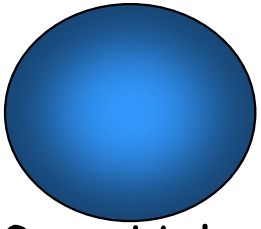
creates and plays with

used for player input and `DotCom` locations

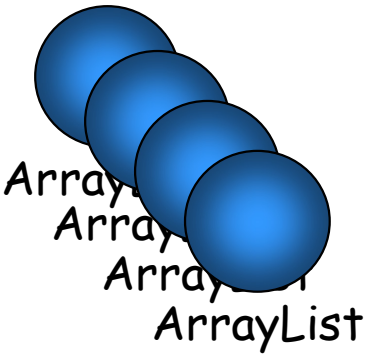
… and the objects







plus 4 ready-baked objects. Instances of ArrayLists are objects too!

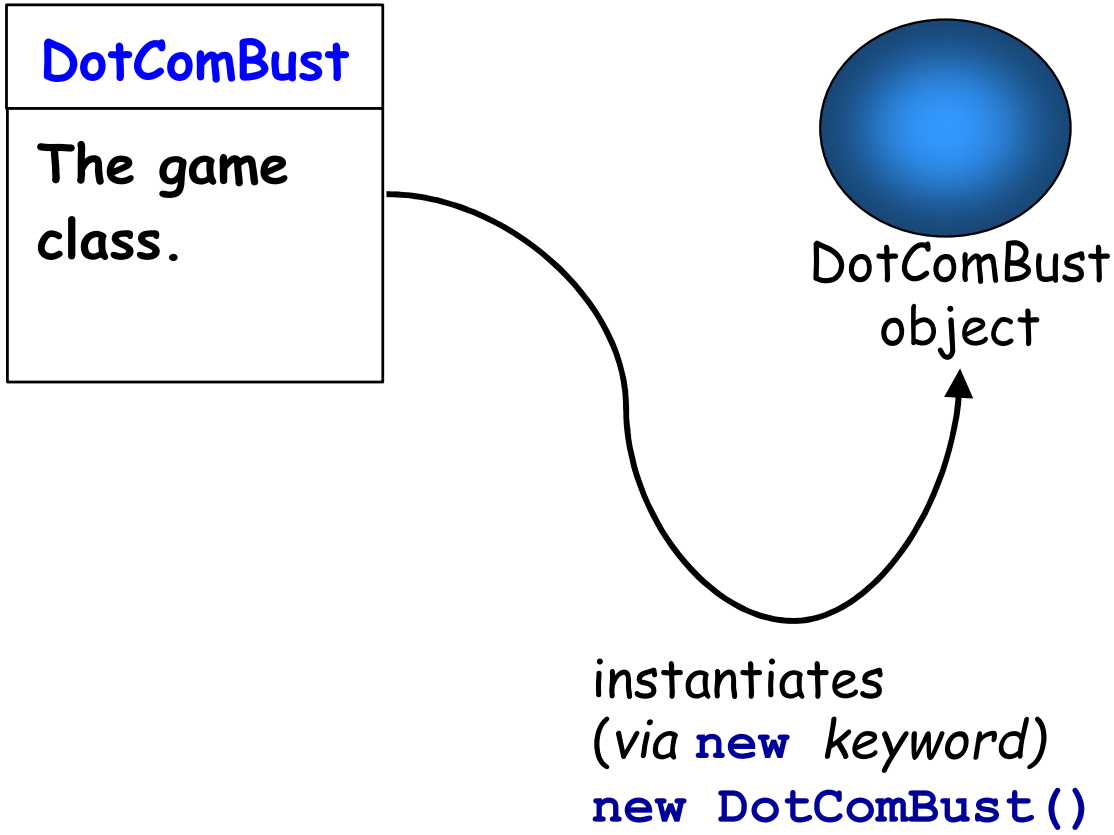


One for DotComBust

One for each instance of DotCom (*three*)

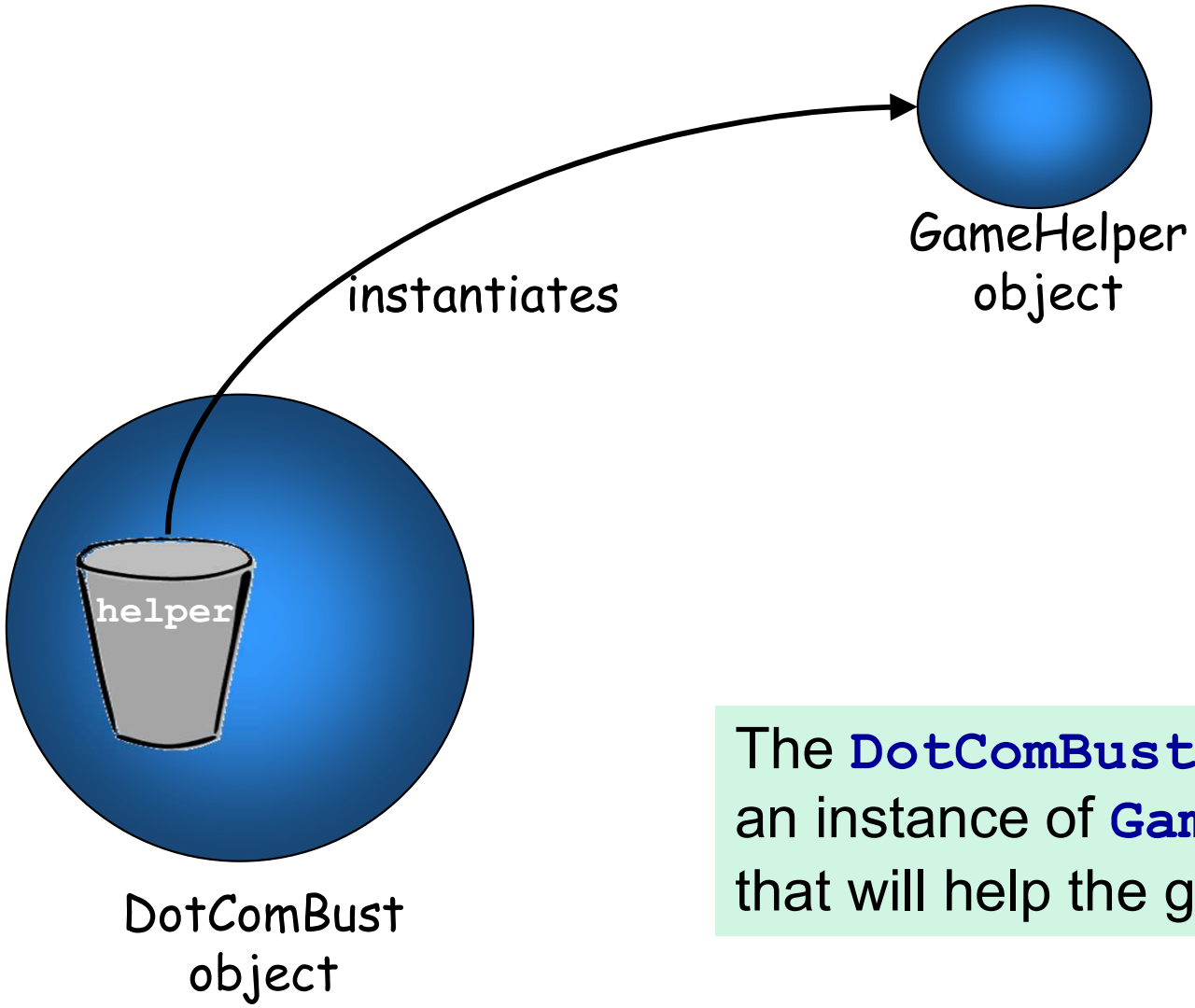
The next slides describe the objects and code necessary to implement a more realistic version of the game, according to page 2.

Who, what, when, where, why?!



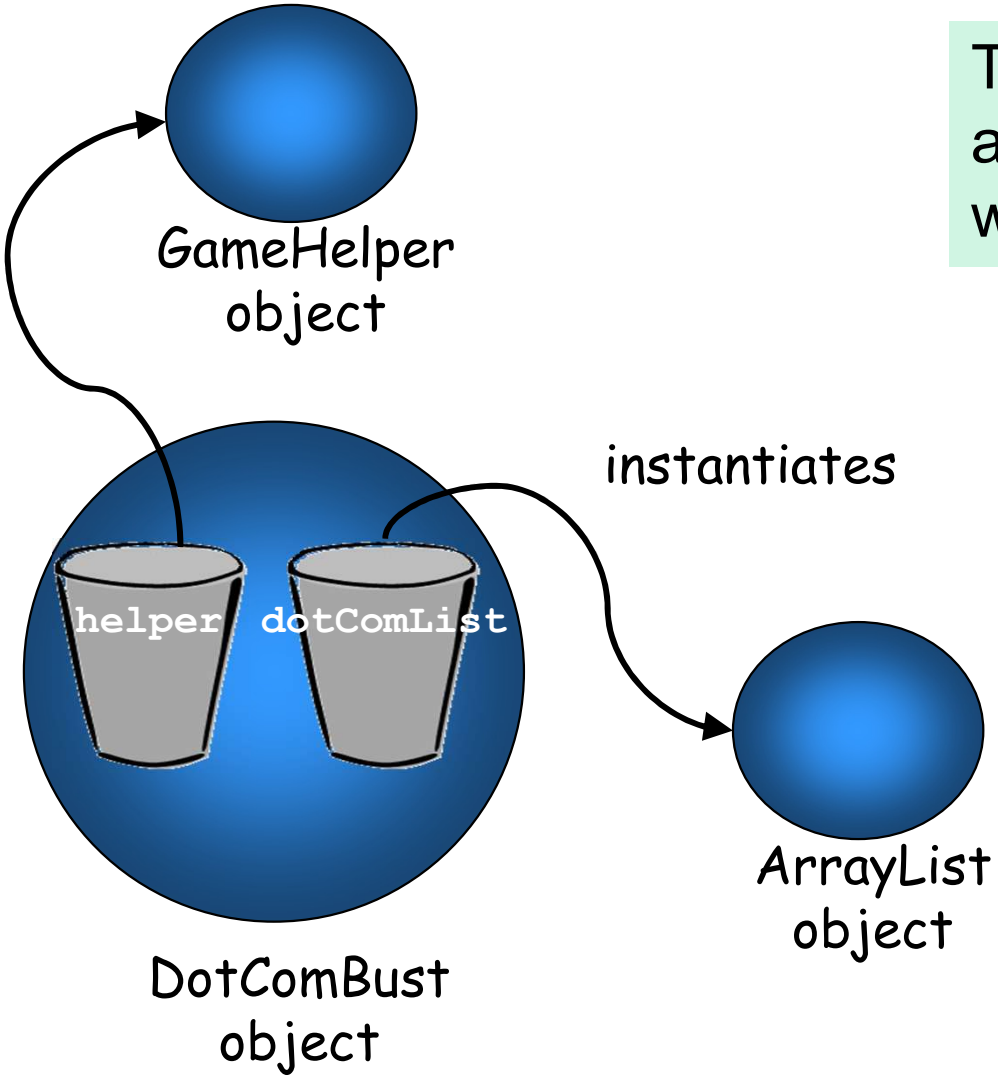
The `main()` method in the `DotComBust` class instantiates the `DotComBust` object that does all the game stuff.

Who, what, when, where, why?! *(cont.)*



The **`DotComBust`** object instantiates an instance of **`GameHelper`**, the object that will help the game do its work.

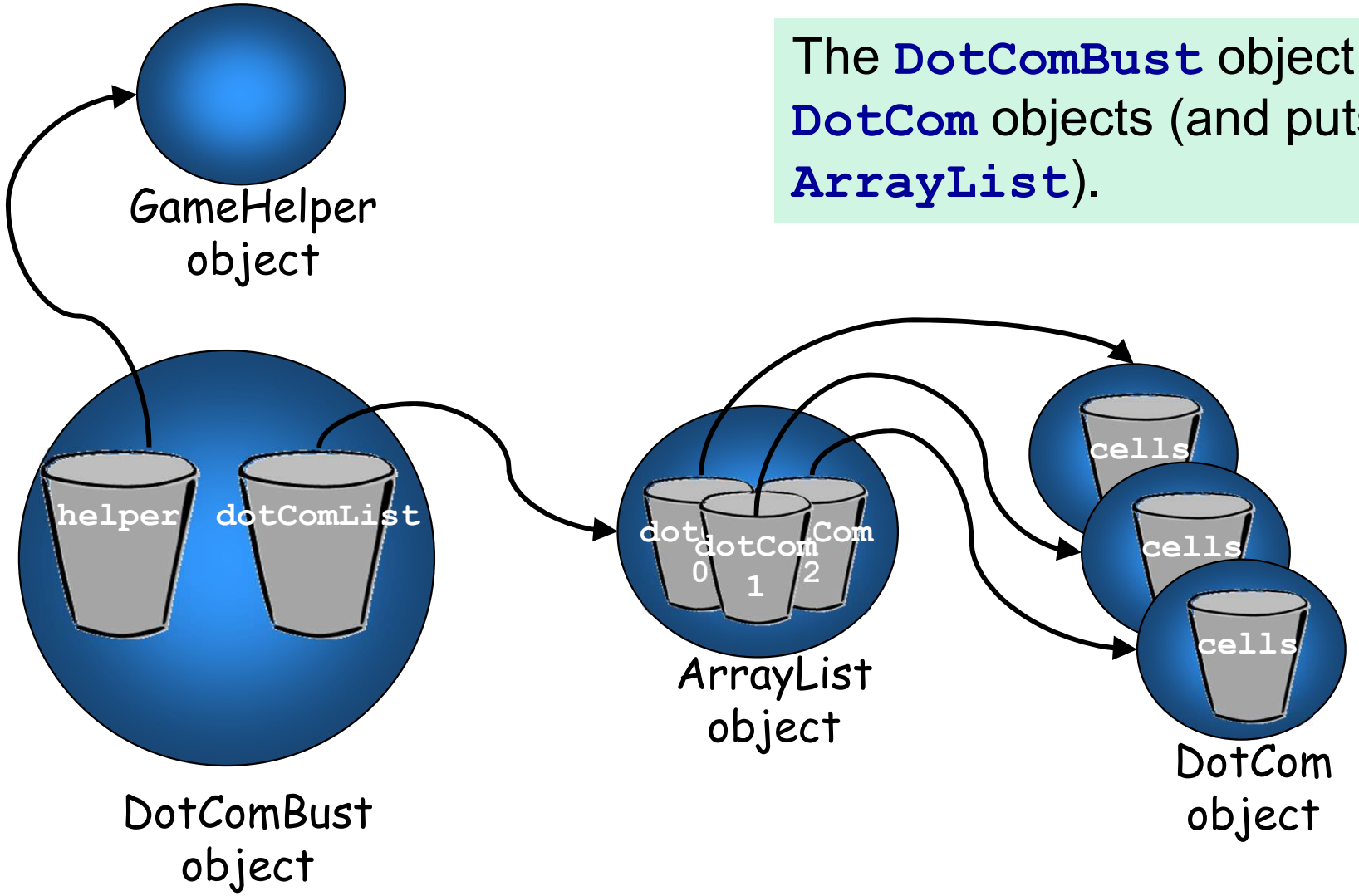
Who, what, when, where, why?! *(cont.)*



The **`DotComBust`** object instantiates an instance of an **`ArrayList`** that will hold the 3 **`DotCom`** objects.

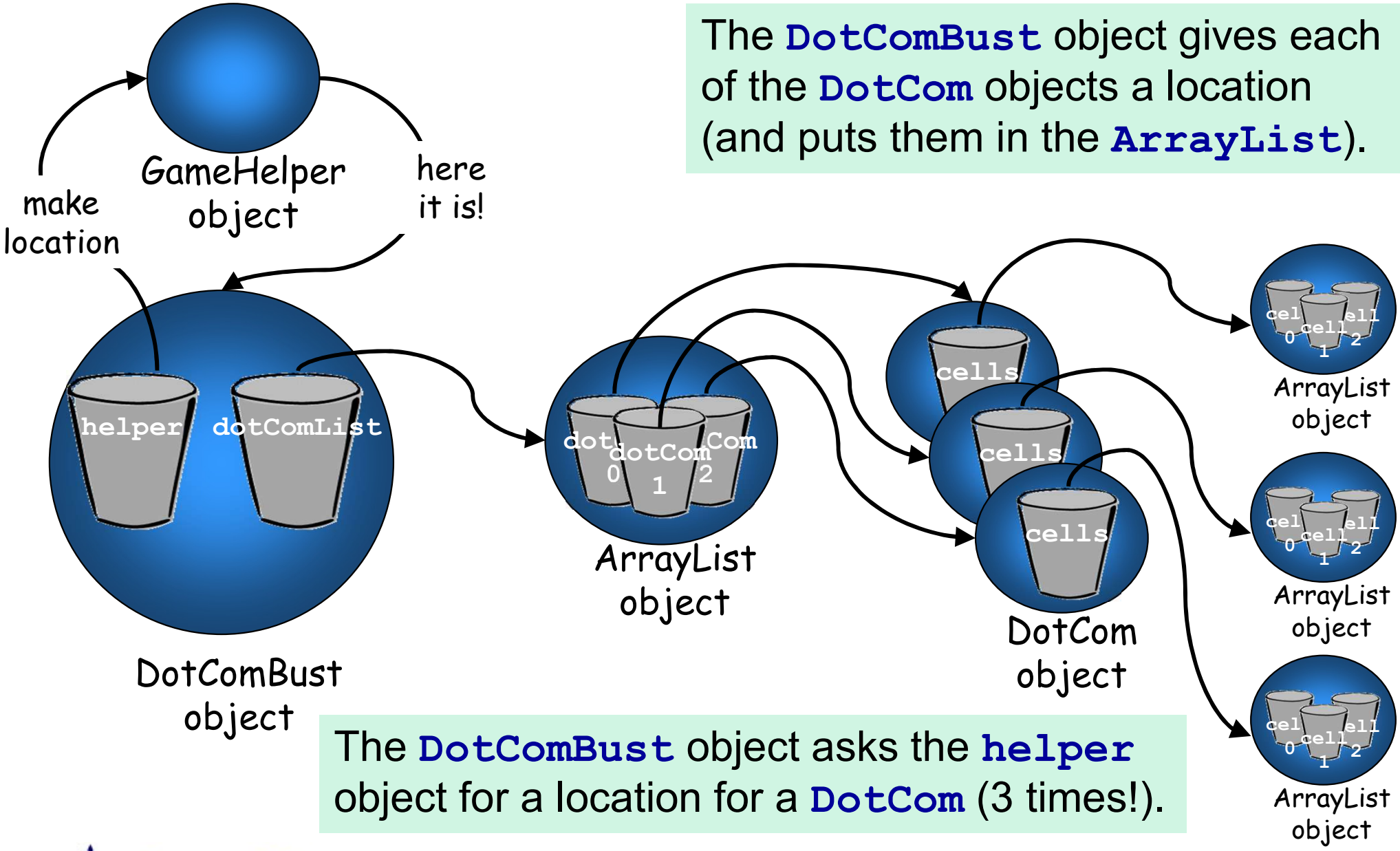
Who, what, when, where, why?! *(cont.)*

The **`DotComBust`** object creates 3 **`DotCom`** objects (and puts them in the **`ArrayList`**).



Who, what, when, where, why?! *(cont.)*

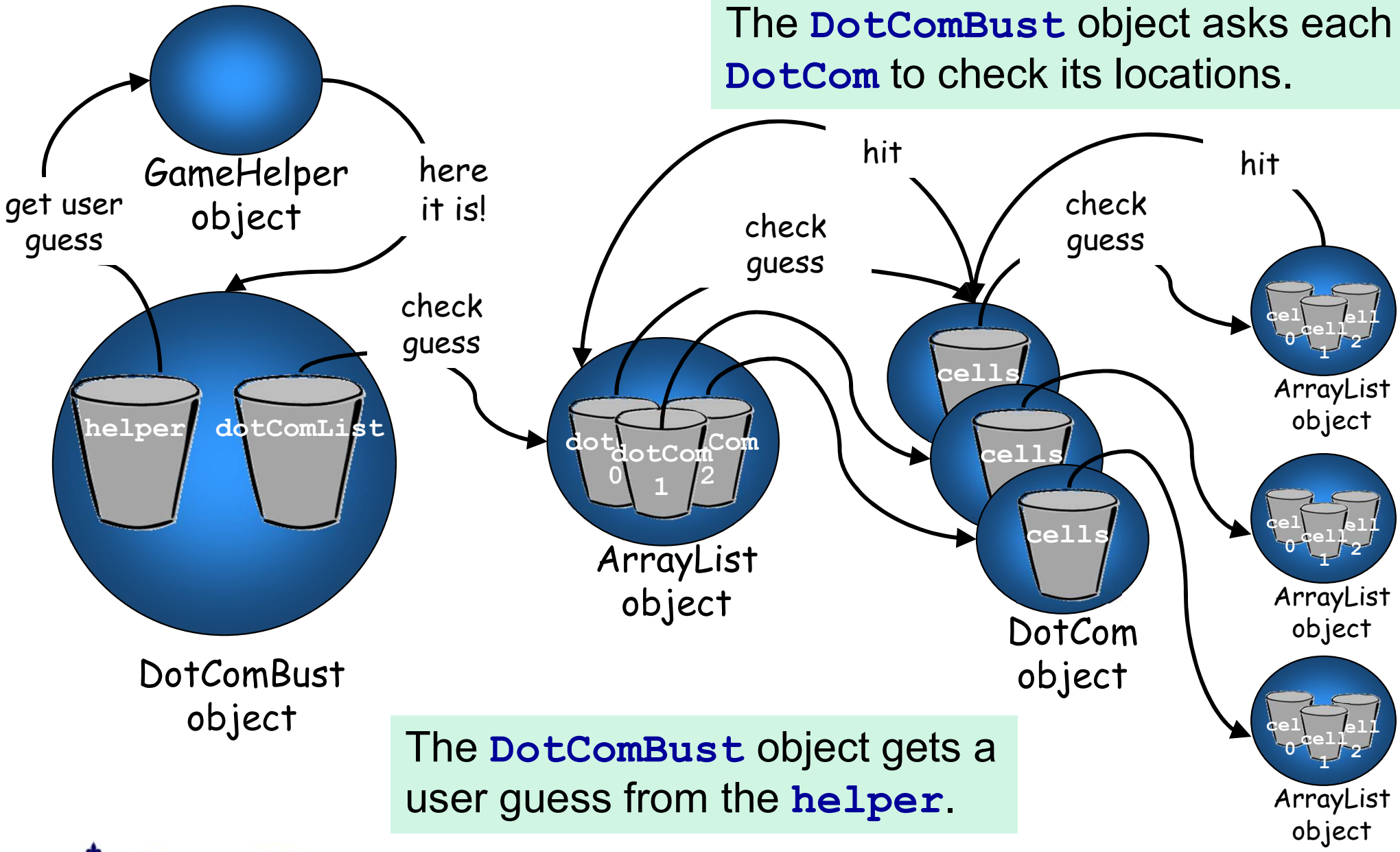
The **DotComBust** object gives each of the **DotCom** objects a location (and puts them in the **ArrayList**).



The **DotComBust** object asks the **helper** object for a location for a **DotCom** (3 times!).

Who, what, when, where, why?! *(cont.)*

The **DotComBust** object asks each **DotCom** to check its locations.



The **DotComBust** object gets a user guess from the **helper**.

The improved code (1/3)

```
import java.util.*;
public class DotComBust {
   private GameHelper helper = new GameHelper();
   private ArrayList<DotCom> dotComList = new ArrayList<DotCom>();
   private int numOfGuesses = 0;

   private void setUpGame() {
     // first make some dot coms and give them locations
     DotCom one = new DotCom();
     one.setName("Pets.com");
     DotCom two = new DotCom();
     two.setName("eToys.com");
     DotCom three = new DotCom();
     three.setName("Go2.com");
     dotComList.add(one);
     dotComList.add(two);
     dotComList.add(three);

     System.out.println("Your goal is to sink three dot coms.");
     System.out.println("Pets.com, eToys.com, Go2.com");
     System.out.println("Try to sink them all in the fewest number
                                 of guesses");
     for (DotCom dc : dotComList) {
       ArrayList<String> newLocation = helper.placeDotCom(3);
       dc.setLocationCells(newLocation);
     }
   }
```

```
for(int i=0; i<dotComList.size(); i++) {
  ArrayList<String> newLocation =
                 helper.placeDotCom(3);
  dotComList.get(i).
      setLocationCells(newLocation);
}
```

The improved code (2/3)

```
private void startPlaying() {
  while(!dotComList.isEmpty()) {
    String userGuess = helper.getUserInput("Enter a guess");
    checkUserGuess(userGuess);
  }
  finishGame();
}

private void checkUserGuess(String userGuess) {
  numOfGuesses++;
  String result = "miss";
  for (DotCom dc : dotComList) {
    result = dc.checkYourself(userGuess);
    if (result.equals("hit")) { break; }
    if (result.equals("kill")) {
      dotComList.remove(dc);
      break;
    }
  }
  System.out.println(result);
}
```

```
for (int i=0; i<dotComList.size(); i++) {
  result = dotComList.get(i).
              checkYourself(userGuess);
}
```

The improved code (3/3)

```
  private void finishGame() {
    System.out.println("All Dot Coms are dead!
                        Your stock is now worthless.");
    if (numOfGuesses <= 18) {
      System.out.println("It only took you "
                          + numOfGuesses + " guesses.");
      System.out.println("You got out before your options sank.");
    }
    else {
      System.out.println("Took you long enough. "
                          + numOfGuesses + " guesses");
      System.out.println(" Fish are dancing with your options.");
    }
  }
  public static void main(String[] args) {
    DotComBust game = new DotComBust();
    game.setUpGame();
    game.startPlaying();
  }
}
```